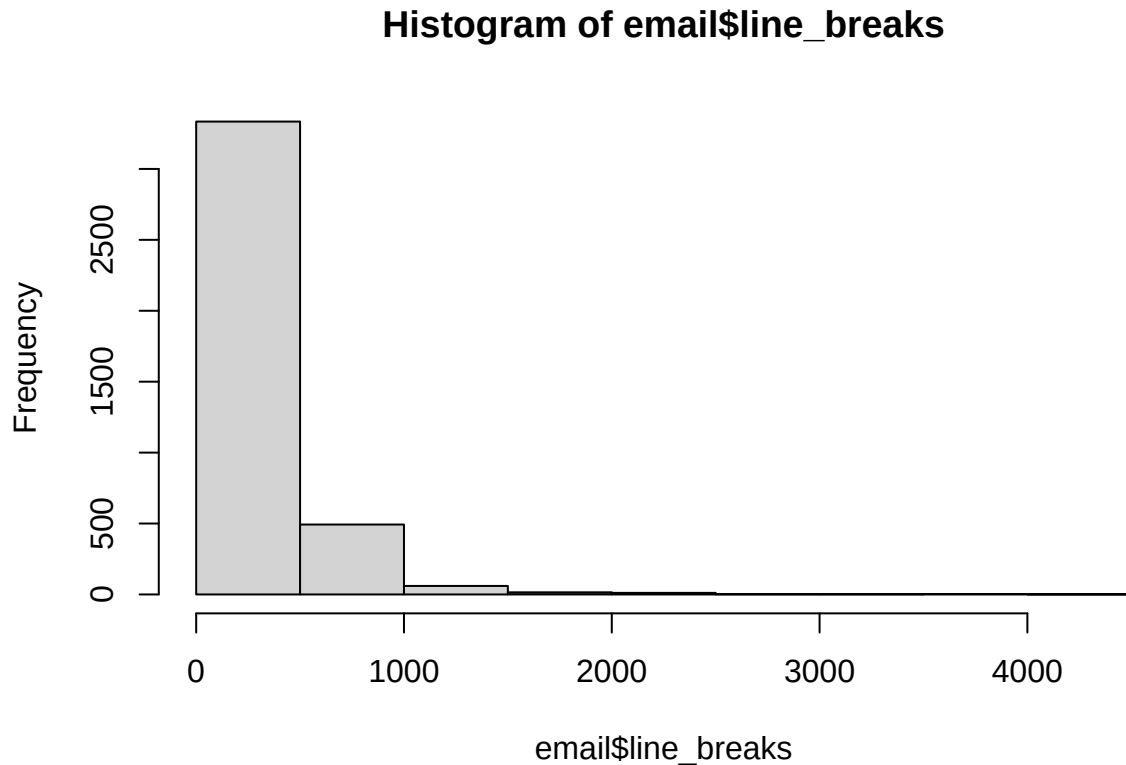


Email Line Breaks

```
email <- read.csv("https://raw.githubusercontent.com/roualdes/data/master/email.csv")
```

A: Plotting

```
hist(email$line_breaks)
```



B: Relevant Variables

```
n <- length(email$line_breaks)
mean <- mean(email$line_breaks)
s <- sd(email$line_breaks)
```

C: Confidence Interval

Two sided 95% CI

```
alpha <- 0.05
stderr <- s / sqrt(n)
tstar <- qt(1 - alpha / 2, n - 1) # two-sided
margin_err <- stderr * tstar

ci <- c(mean - margin_err, mean + margin_err)
ci
```

```
## [1] 220.6611 240.6560
```

D: Hypothesis Test

$H_0: \mu = 250$

$H_1: \mu \neq 250$

```
mu <- 250
t <- (mean - mu) / stderr
t
```

```
## [1] -3.793
```

```
pvalue <- 2 * (1 - pt(abs(t), n - 1))
pvalue
```

```
## [1] 0.0001510817
```

```
pvalue < alpha
```

```
## [1] TRUE
```

E: Conclusion

Reject H_0 at the 5% level.

F: Practical Conclusion

There is strong evidence that the population mean number of line breaks is not 250.